Tutorial 1

1. Traditionally, computers use binary arithmetic, and store n-bit words composed of zeros and ones. The new m-computer uses m-ary arithmetic, and stores n-symbol words in which the symbols (m-ary digits) are from the set $\{0, 1, \ldots, m-1\}$. How many n-symbol words are there with k_0 zeros, k_1 ones, k_2 twos, ..., k_{m-1} copies of symbol $m-1$, where $k_0+k_1+k_2+...+k_{m-1} = n$?

2. The following is taken from Gubner's "Probability and Random Processes for Electrical and Computer Engineers".

Signal processing. In a radar system, the voltage of a noise waveform at time t can be viewed as possibly being any real number. The first step in *modeling* such a noise voltage is to consider the sample space consisting of all real numbers, i.e., $\Omega = (-\infty, \infty)$.

Computer memories. Suppose we store an n-bit word consisting of all 0s at a particular location. When we read it back, we may not get all 0s. In fact, any n-bit word may be read out if the memory location is faulty. The set of all possible n-bit words can be modeled by the sample space

$$\Omega = \{(b_1, \ldots, b_n) : b_i = 0 \text{ or } 1\}.$$

Optical communication systems. Since the output of a photodetector is a random number of photoelectrons. The logical sample space here is the nonnegative integers,

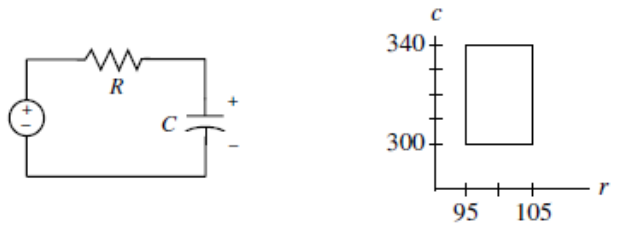
$$\Omega = \{0, 1, 2, \ldots\}.$$

Notice that we include 0 to account for the possibility that no photoelectrons are observed.

Wireless communication systems. Noncoherent receivers measure the energy of the incoming waveform. Since energy is a nonnegative quantity, we model it with the sample space consisting of the nonnegative real numbers, $\Omega = [0, \infty)$.

Variability in electronic circuits. Consider the lowpass RC filter shown in Figure 1.5(a). Suppose that the exact values of R and C are not perfectly controlled by the manufacturing process, but are known to satisfy

$$95 \text{ ohms} \le R \le 105 \text{ ohms} \quad \text{and} \quad 300\,\mu\text{F} \le C \le 340\,\mu\text{F}.$$



(a) (b)

Figure 1.5. (a) Lowpass RC filter. (b) Sample space for possible values of R and C.

This suggests that we use the sample space of ordered pairs of real numbers, (r, c), where $95 \le r \le 105$ and $300 \le c \le 340$. Symbolically, we write

$$\Omega = \{(r, c) : 95 \le r \le 105 \text{ and } 300 \le c \le 340\},$$

which is the rectangular region in Figure 1.5(b).

Computer network traffic. If a router has a buffer that can store up to 70 packets, and we want to model the actual number of packets waiting for transmission, we use the sample space

$$\Omega = \{0, 1, 2, \ldots, 70\}.$$

Notice that we include 0 to account for the possibility that there are no packets waiting to be sent.

(2.12)

12. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class, and 16 in the German class. There are 12 students who are in both Spanish and French, 4 who are in both Spanish and German, and 6 who are in both French and German. In addition, there are 2 students taking all 3 classes.

(a) If a student is chosen randomly, what is the probability that he or she is not in any of the language classes?

(b) If a student is chosen randomly, what is the probability that he or she is taking exactly one language class?

(c) If 2 students are chosen randomly, what is the probability that at least 1 is taking a language class?

(2.37)

37. An instructor gives her class a set of 10 problems with the information that the final exam will consist of a random selection of 5 of them. If a student has figured out how to do 7 of the problems, what is the probability that he or she will answer correctly

(a) all 5 problems?

(b) at least 4 of the problems?

(2.46)

46. How many people have to be in a room in order that the probability that at least two of them celebrate their birthday in the same month is at least $\frac{1}{2}$? Assume that all possible monthly outcomes are equally likely.

55. Compute the probability that a hand of 13 cards contains

(a) the ace and king of at least one suit;

(b) all 4 of at least 1 of the 13 denominations.